Fiscal Year:	FY 2022	**Task Last Updated:**	FY 04/13/2026
PI Name:	Karma, Alain Ph.D.		
Project Title:	Effects of Curvature and Unsteady Dynamics on Alloy Solidification Patterns (CUSP)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	a.karma@neu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617 373 2929
Organization Name:	Northeastern University		
PI Address 1:	Department of Physics		
PI Address 2:	360 Huntington Ave		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02115-5005	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	02/08/2022	**End Date:**	02/07/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Strutzenberg Littles, Louise	**Contact Phone:**	(256) 544-0946
Contact Email:	louise.s@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Erb, Randall (Northeastern University)		
Grant/Contract No.:	80NSSC22K0664		
Performance Goal No.:			
Performance Goal Text:			
	During the manufacturing of industrial components by solidification processes that make use of casting welding and additive manufacturing, the properties and the reliability of the finished products depend on the microstructure. The microstructure that is formed is controlled by the alloy composition and the processing conditions of growth rate and thermal gradient at the solidifying interface. Quantitatively relating the detailed microstructural features with the growth is crucial for selecting alloy composition and designing growth conditions to obtain microstructure with optimum properties in the finished product. From a fundamental standpoint, the formation of different solidification microstructures is controlled by a subtle interplay between microscopic interfacial phenomena and macroscopic transport of heat and solute, which is strongly influenced by gravity-induced convection in the liquid under terrestrial conditions. Therefore, directional solidification experiments are conducted on the International Space Station (ISS) to investigate dynamic microstructure development under diffusive growth and to obtain benchmark data in three-dimensional (3D) bulk samples. The unique in situ optical observations of transparent organic alloys in the Dispositif d'Etude de la		

Task Description:	Croissance et des LIquides Critiques (DECLIC) Directional Solidification Insert (DSI) make it possible to gain fundamental insights into mechanisms of microstructure formation by quantitative computational modeling of those observations. The objective of the CUSP project is to address the effect macroscopic curvature and unsteady dynamics have on the solidification front through modeling and ground-based solidification studies. Microstructure formation has been extensively studied under steady-state growth conditions where the macroscopic envelope of the solidification front is assumed to be planar, and the growth rate is constant. Under those conditions, the interface evolves primarily with a stationary growth shape. However, in most industrial solidification processes, the macroscopic envelope of the solidification front is generally curved on a larger length scale than the pattern scale corresponding to the primary cellular or dendritic array spacing. Latent heat generated at the solid-liquid interface generally induces a radial temperature gradient in directional solidification of bulk samples, which in turn produces a macroscopic curvature of the solidification front. Processing methods such as welding and additive manufacturing also inherently involve large scale front curvature associated with the shape of the melt pool. Unsteady dynamics are found after changes in growth conditions or in dendrites with side-branching close to the tip. The growth rate can change continuously or abruptly during solidification processing to generate spatially graded microstructures. This is true in casting (where the solidification rate decreases from the surface of the mold to the interior of the casting), and in additive manufacturing (where the growth rate changes by orders of magnitude from the bottom to the top of the melt pool). Both of these effects are further investigated by the in situ observations of experiments in transparent organic succinonitrile (SCN)-camphor alloys of different compositions in the DSI-R1 campaign, and are planned for the DSI-R2 campaign.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New Project for FY2022. Project entered into the NASA Task Book in April 2026 (Ed., 4/13/26).
Bibliography Type:	Description: (Last Updated: 05/10/2026)